

AICTE Productization Fellowship YUKTI Innovation Challenge 2025 Institution's Innovation Council (IIC), AICTE

2nd Stage Evaluation

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VERBOFIX

Unmasking Silence: Advancing Dysfluent Speech Understanding with Next-
Generation Tools

Assignment 1: Problem Framing Canvas



Idea



Business Context



Customer Context

Our idea, solution, hypothesis, business need, challenge ...

Our idea is to develop **Verbofix**, a speech-fluency support system that analyzes a user's recorded speech to identify stammering patterns and convert them into a clear severity score with a simple progress timeline.

Present

What is the current state of this idea, solution, challenge, project?
(critical situation, part of a new initiative, new business strategy, etc.)

Core functionality—audio input, transcription pipeline, and stammer pattern detection—is in early working prototype form, with planned refinements for accuracy and performance.

Progress timeline visualization, severity scoring, and recommendation engine are partially implemented and undergoing testing.

Initial feedback gathered from students and peers suggests strong interest in quick, objective speech-feedback tools, indicating real demand.

The project currently relies on limited publicly available datasets and self-collected samples; broader, diverse data is needed to improve robustness and reduce bias.

The system has not undergone therapist-level validation, which is essential for credibility in speech pathology environments.

Current stack (Flask backend, React frontend, Whisper-based transcription) is functioning well for MVP development.

Potential user segments (schools, clinics, students) have been identified, but pricing and business model testing are yet to begin.

Surveys and discussions with students, job seekers, and teachers highlight the need for accessible stammering support tools, proving market relevance.

Who are the customers, users, target audience in need of this solution?
Who should we focus on first?

Students (school/college), job seekers, young professionals, individuals with mild to moderate stammering.

Speech therapists, special educators, clinical practitioners, counseling centers.

Parents of children with early speech fluency issues and teachers identifying communication challenges in classrooms.

When/Where do they need our solution?
When do they struggle the most?

Before interviews, presentations, or public speaking tasks where fluency impacts confidence and performance.

In therapy sessions, to assist therapists with quick analysis and standardized scoring. In rural/semi-urban areas, where access to speech specialists is very limited.

How are they solving their problems today?
What behaviors can we see?

Visiting speech therapists occasionally, relying on manual, subjective assessments.

Using general communication apps that focus on pronunciation, not stammering.

Users repeatedly record their voice but cannot identify problems on their own.

They lose practice consistency because there is no system to track progress.

Past

How we (others) tried to solve it in the past?

What worked well? What were some small wins?

Traditional speech therapy sessions helped users improve through structured exercises, repetition training, and face-to-face feedback.

Some mobile apps focused on communication or pronunciation offered practice routines and confidence-building exercises that helped users stay motivated.

Basic tele-therapy sessions improved accessibility by reducing travel barriers and enabling remote consultations.

Therapist-led progress notes allowed some level of month-to-month tracking, even though not standardized.

What didn't work so well? Why didn't we succeed?

Manual assessments were slow and inconsistent, making it impossible to provide instant feedback or track progress daily.

Most apps focused on pronunciation, not stammering, leaving a major gap in tools for detecting difficulties like blocks, repetitions, or prolongations.

Therapy follow-up was often irregular, causing users to lose momentum due to lack of accountability and continuous guidance.

No standardized scoring system existed, so different therapists measured progress differently, confusing users and reducing trust.

Early self-practice tools lacked analytics, meaning users didn't know whether they were improving or repeating the same mistakes.

Most digital tools ignored real-world recording conditions, causing inaccurate analysis in noisy environments.

What has the target audience tried in the past?

What worked well? What were some enablers, alternative solutions?

Regular speech therapy sessions helped users improve when guided by trained experts through structured exercises and repetition practice.

Self-recording and listening back to their own speech increased awareness of stammering patterns and motivated some users to practice more consistently.

What didn't work so well? What were some obstacles?

General communication apps were not designed for stammer detection, leaving users without targeted support or measurable improvement tracking.

Manual assessment was slow, subjective, and inconsistent, making it difficult for users to get frequent or accurate feedback on their progress.

Future

What are our competitors doing about it?

AI speech tools (e.g., Pronunciation apps) analyze clarity, not **disfluency patterns** like blocks or repetitions.

Some therapist platforms offer tele-sessions but lack **automated progress tracking**, requiring manual evaluation.

No major platform currently provides an **audio timeline visualization** of stammering incidents, leaving a big opportunity gap.

A few global speech therapy apps (like **Stamora**, **Speekee**, **ELSA**) provide practice routines, but none offer **objective stammer event detection or severity scoring**.

What are some trends working against/in favor of this idea?

High smartphone and internet penetration makes digital fluency tools accessible even in rural regions.

Growing acceptance of AI in healthcare increases trust in automated assessments.

Tele-therapy adoption is rising, creating demand for AI-assisted diagnostic tools.

Government digital health initiatives support AI-enabled screening and remote care.

Where are some opportunities?

Become the **first widely adopted digital stammer-analysis system** with automated severity scoring and progress tracking.

Partner with **schools, clinics, special educators, and corporates** for large-scale adoption.

Expand to **multi-language support** and dominate underserved Indian language markets.

What are some trends and changes that might influence our customers/users/target audience?

Increasing screen time and reduced in-person communication make speech confidence issues more common among students.

Speech therapists are overloaded, creating long waiting times and inconsistent follow-up for users.

If nothing changes, how would the future look like for them?

Users will continue struggling with low confidence during interviews, presentations, and social interactions.

Growing acceptance of digital learning and tele-therapy encourages users to seek online fluency tools.

AI-based assessment tools are becoming more trusted, making automated speech analysis more acceptable.

They will miss the chance to receive early, data-driven intervention that could significantly improve fluency.

Needs/Goals

What do we want to achieve?

(The desired outcome, the pain we want to solve, the impact on our business, KPIs.)

Create a reliable and accessible system for early identification and monitoring of stammering patterns.

Standardize progress tracking with measurable scores and timelines across users and clinics.

Enable users to receive instant, data-driven feedback that accelerates speech improvement.

Achieve high user engagement and measurable improvement rates as key performance indicators.



Problem Statement

Our (customer, user, target audience)

students, job seekers, young professionals, and individuals with speech fluency difficulties

has the problem (issue, challenge, pain)

of not knowing the severity of their stammering, lacking consistent feedback, and being unable to track whether their speech is improving

when (the situation, context, process step)

they practice alone without expert guidance, prepare for interviews or presentations, or have limited access to speech therapists

Our solution will (customer value, benefit for the customer)

provide instant speech analysis, objective severity scoring, clear progress timelines, and personalized exercises that help them confidently improve their fluency

and also help our business (value, benefit for the business)

by creating a scalable digital therapy support system that can be adopted by schools, clinics, and corporates through subscription, licensing, and partnership channels

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Problem Statement Canvas

CONTEXT
When does the problem occur?

Your Answer:
The problem occurs when individuals attempt to speak in situations that demand fluency—such as interviews, presentations, classroom interactions, or spontaneous conversations—without access to consistent feedback or professional guidance. It becomes worse when practice happens alone, in stressful scenarios, or when therapy follow-up is irregular.

PROBLEM
What is the root cause of the problem?

Your Answer:
The root cause is the lack of an objective, accessible, and continuous method to detect stammering patterns and measure improvement. Manual assessments are slow and inconsistent, therapy availability is limited, and users cannot clearly understand their progress without expert input.

ALTERNATIVES
What do customers do now to fix the problem?

Your Answer:
Customers either visit speech therapists occasionally, record their own voice for self-analysis, rely on generic communication apps, or search for online exercises. Some simply avoid speaking situations and hope practice alone will help.

CUSTOMERS
Who has the problem most often?

Your Answer:
Students, job seekers, young professionals, and individuals with speech fluency difficulties—especially those in rural and semi-urban regions with limited access to certified therapists. Parents and teachers also face difficulty identifying early signs in children.

EMOTIONAL IMPACT
How does the customer feel?

Your Answer:
Customers feel anxious, self-conscious, and frustrated because they cannot understand why their speech isn't improving.

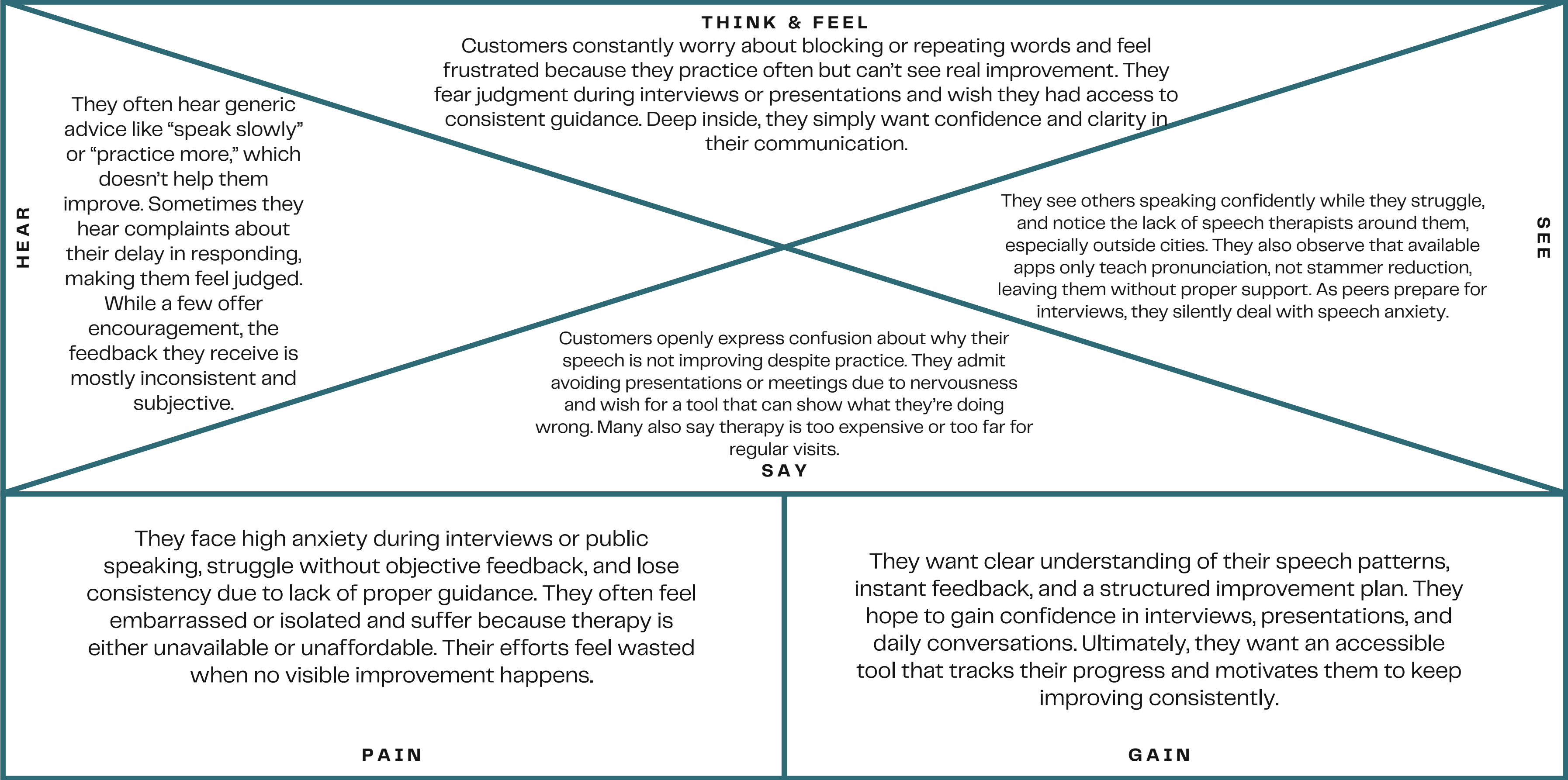
QUANTIFIABLE IMPACT
What is the measurable impact (include units)?

Your Answer:
Stammering affects academic and job outcomes: reduced interview success rates, lower participation in presentations, and significant loss in communication effectiveness.

ALTERNATIVE SHORTCOMINGS
What are the disadvantages of the alternatives?

Your Answer:
Manual assessments lack consistency, general communication apps do not measure stammering, and self-recording provides no analysis or guidance. Therapy is expensive, limited in frequency, and not scalable. Without a structured system, users cannot track progress or maintain steady improvement.

Assignment 2: Customer Empathy Map

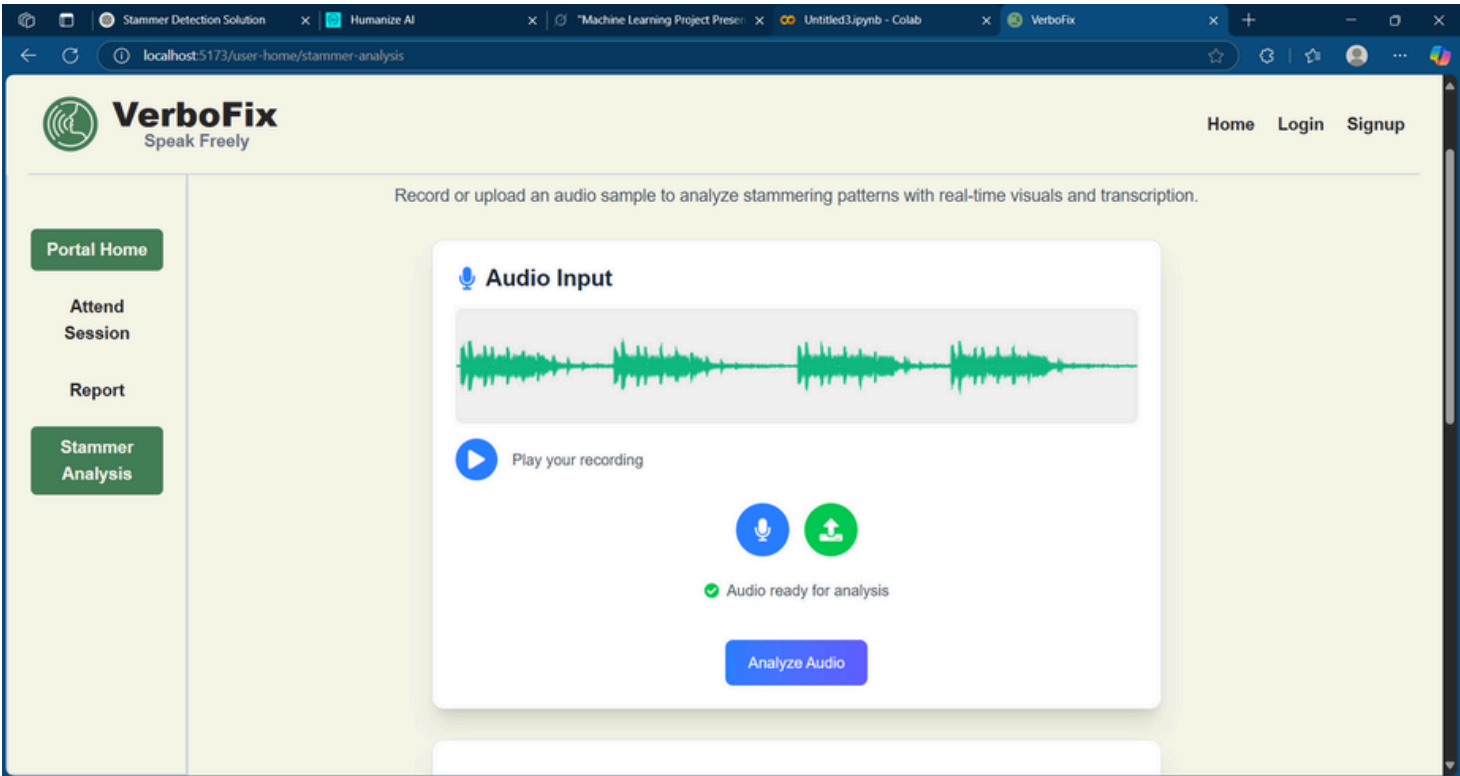
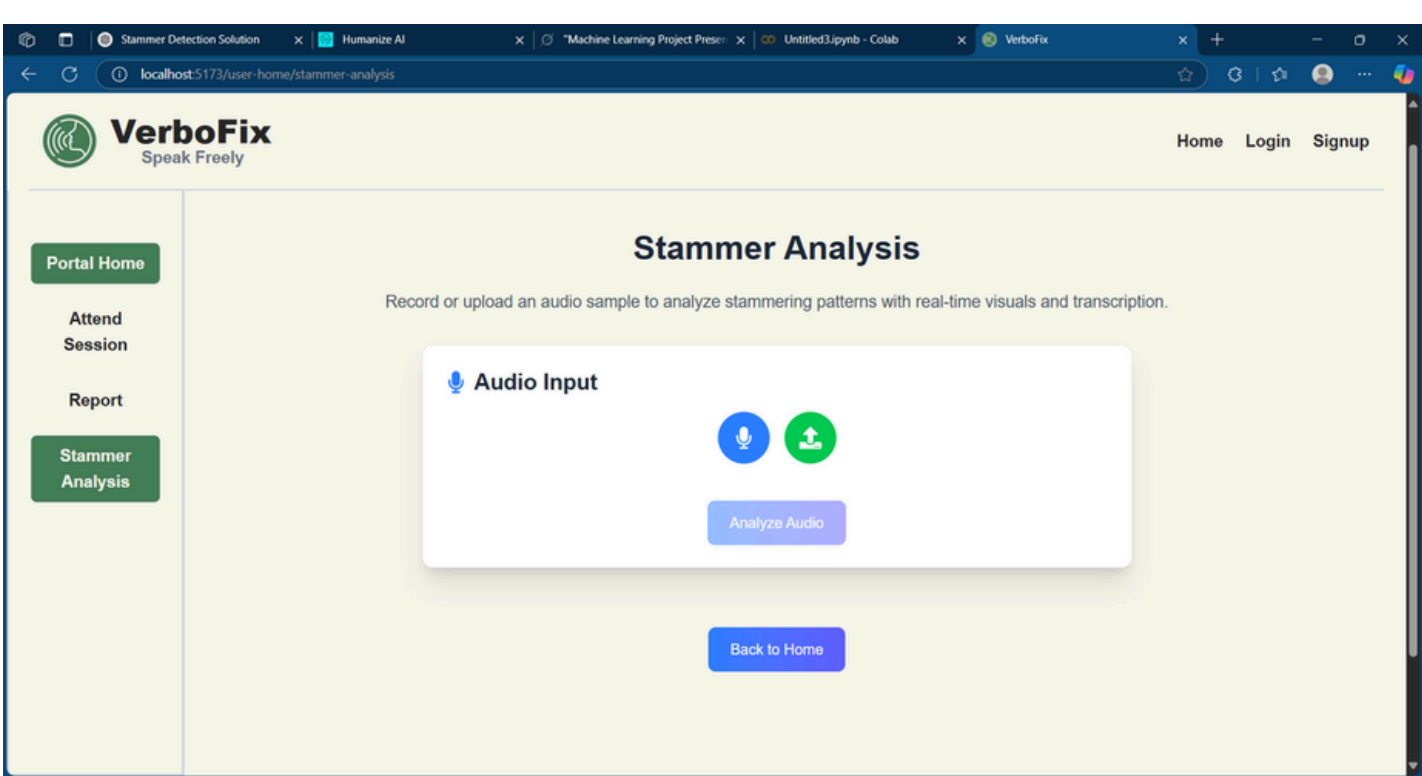


Assignment 3: Business Model Canvas

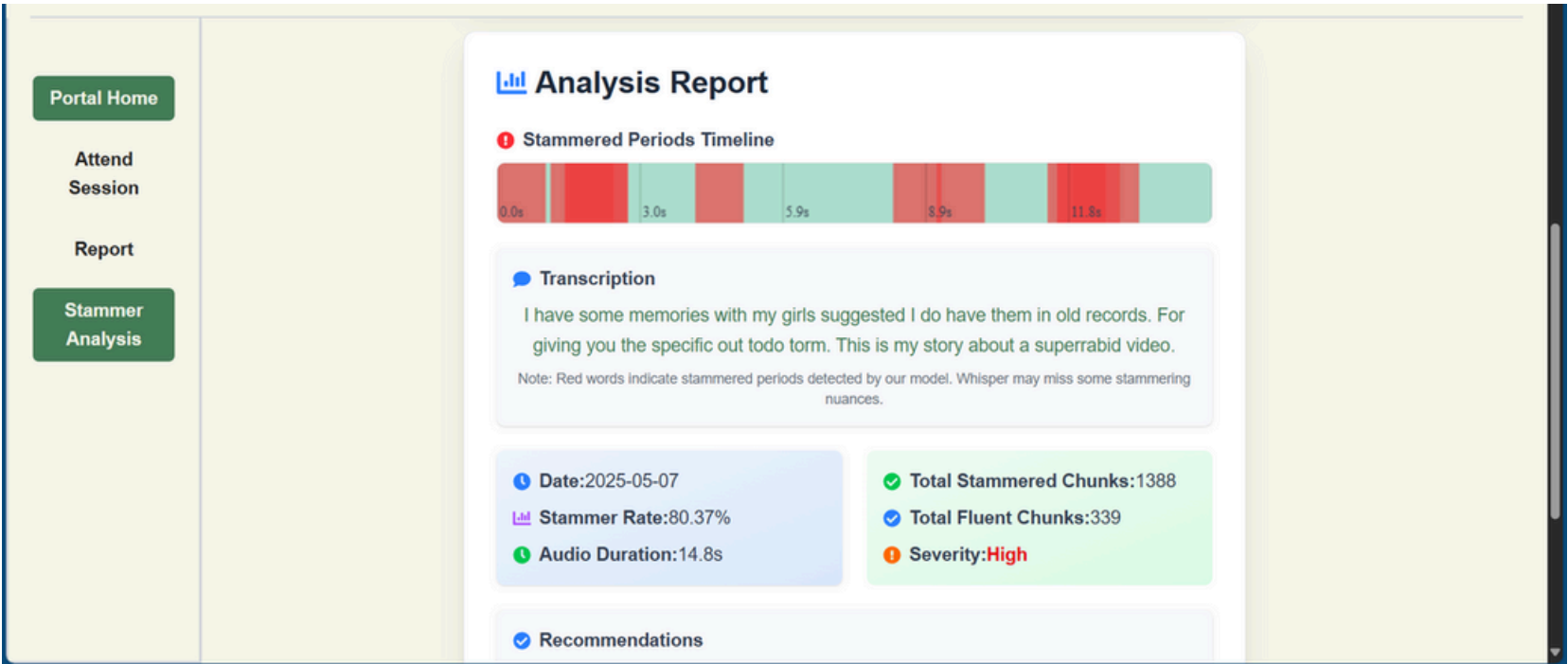
Key Partners • Speech therapy clinics & rehabilitation centers • NGOs supporting children with disabilities • Special education schools & institutions • Incubation Support Teams	Key Activities • Dysfluency detection • Real-time testing with dysfluent speakers	Value Propositions • Inclusive ASR system that understands dysfluent speech • Combined Transcription + Detection • Enables accessibility for future AI teachers & virtual assistants	Customer Relationship • Long-term partnerships with schools & clinics • Subscription-based usage support	Customer Segments • Students with speech disorders • Speech therapy clinics & hospitals • Virtual assistant developers
	Key Resources • Custom dysfluent speech dataset • Annotation tools & phoneme-level labeling		Channel • Direct outreach to schools & therapy centers • Mobile & web applications	
Cost Structure • GPU training costs for Whisper fine-tuning • Cloud storage + server deployment • Speech therapist consultation costs			Revenue Stream • ASR + detection API subscription • Custom deployments for therapy centers • Software/communication licensing	

Assignment 4: Images of Prototype/Innovation

Communication System



ANALYSIS REPORT



Assignment 5: Presentation

1. The Overview

A very high-level overview of our product or service

We are a student-led research team developing Verbofix, a real-time, AI-powered stammering detection and inclusive speech recognition system designed to assist individuals with speech disfluencies, therapists, and accessibility platforms.

Our goal is to turn Verbofix into an affordable, deployable speech-therapy tool that improves communication confidence for people who stammer.

2. The Problem

What problem are we trying to solve? Is it really a problem?

Speech disfluencies such as stammering disrupt communication, causing hesitation, repetition, and blocks that impact daily interactions, interviews, presentations, and social confidence.



Existing speech-recognition tools are trained mostly on fluent speech, making them unreliable and biased for people with stammering.



Despite advances in AI, mainstream ASR systems (Siri, Alexa, Google Assistant) perform poorly on dysfluent speech, leading to mis-transcriptions, dropped words, and user exclusion.

Over **80 million** people globally are affected, yet no real-time, lightweight, clinically interpretable stammer detection solution exists today.

Accurate, instant feedback is essential for therapy, accessibility, and inclusive digital communication—**but current technology fails to provide it.**

3.The Solution

How we are solving the problem

Detect

A lightweight **1-D Convolutional Neural Network** identifies stammered segments in real time by analyzing **MFCCs, delta-MFCCs, energy, pitch, and pause** patterns.

Analyze

A transparent, reproducible feature-fusion pipeline processes speech into (86×29) matrices, enabling interpretable insights into **repetitions, prolongations, and blocks** essential for therapy and accessibility.

Support

Integrated into a **Flask backend and React frontend**, Verbofix provides **instant visual feedback**, live recording, and detailed disfluency detection to support users, therapists, and ASR applications.

4. The Product/Service

Product Working Flow



Data Collection :

Uses professionally annotated stammering datasets (e.g., UCLASS) to build reliable speech samples. Audio is segmented into 1-second chunks.

Lightweight CNN Classifier :

A compact Conv1D model designed for low-latency inference, capable of identifying dysfluent speech segments in real time.

Feature Extraction Engine :

Transforms raw audio into MFCCs, delta-MFCCs, pitch, energy, and pause-duration features, capturing the acoustic signatures of disfluency.

Assistive Speech Insights :

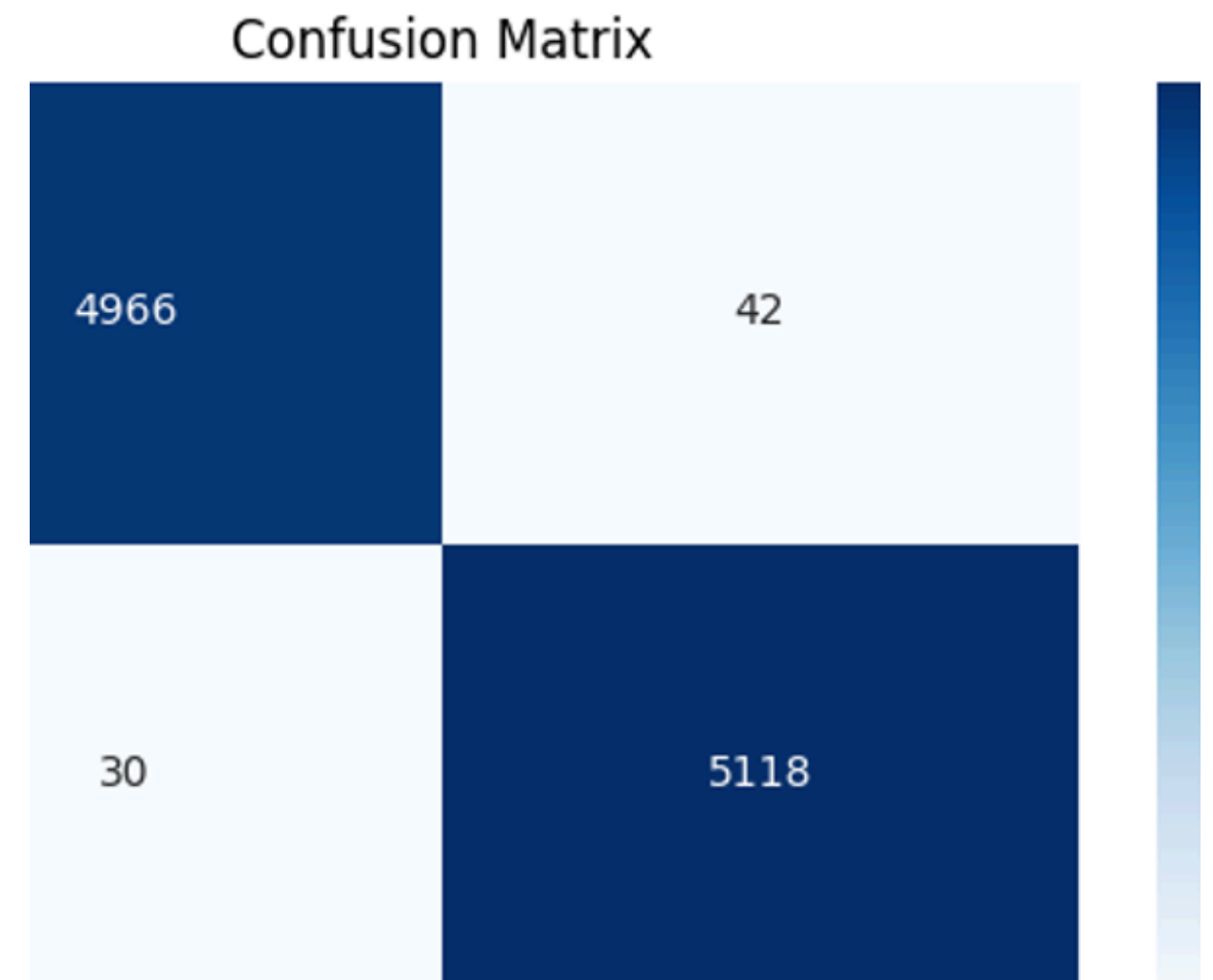
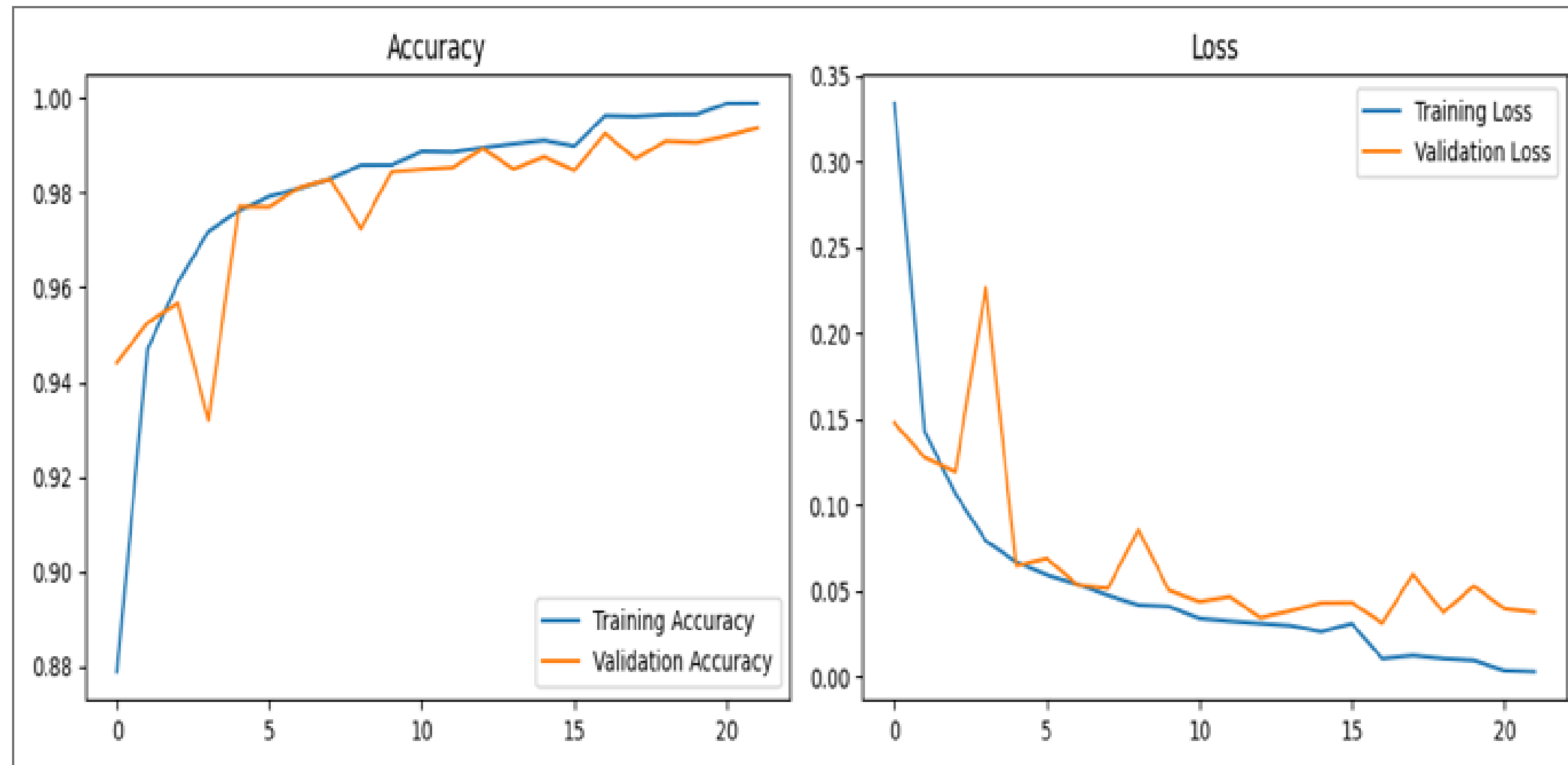
Outputs interpretable fluency metrics and disfluency markers that support self-practice, therapy sessions, and accessible communication tools.

Interactive Web Application :

A Flask backend and React interface allow users to record audio, upload speech samples, and view live stammer detection results with clear visual feedback.

4. The Product/Service

Machine Learning Classification



DATA & DEVELOPMENT

5. Market & Opportunity

The size of our target market

TAM (Global ASR & Voice Technology Market):
~\$30B+ (growing at ~18–20% CAGR)

SAM (Assistive Technology + EdTech + Speech Therapy Market in India):

Schools, EdTech platforms, speech therapy clinics

Target Market (annual potential):

**schools adopting AI-based learning
EdTech companies**

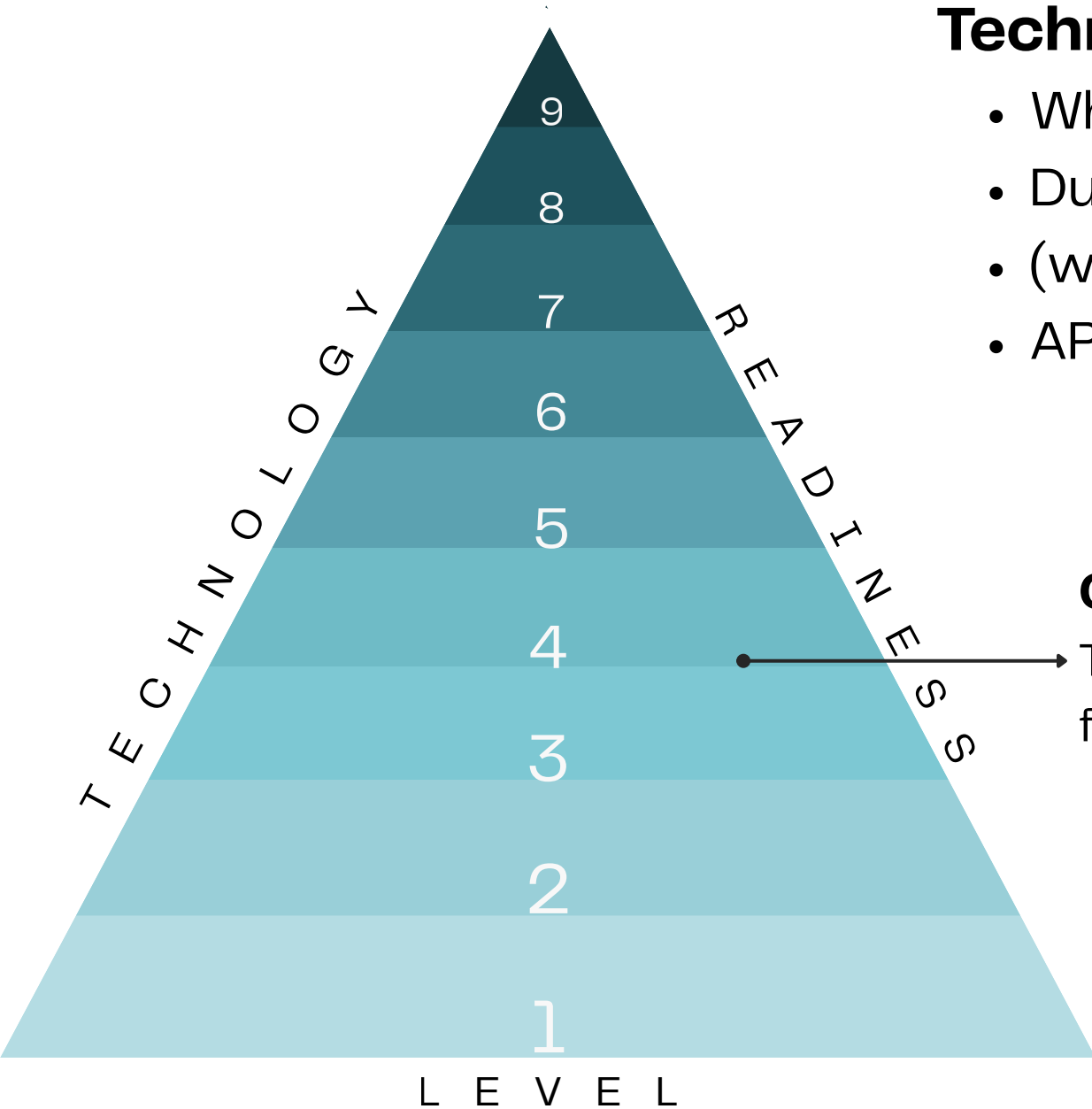
**Speech therapy clinics and hospitals
NGOs supporting children with disabilities**

Market Insight:

There is a growing need for inclusive ASR tools that accurately understand dysfluent speech. As AI teachers, virtual assistants, and voice-driven applications become mainstream, systems must become accessible to people with speech disorders

6. Technology/Innovation

The technologies we're using to solve the problem



Technologies Used:

- Whisper-based ASR fine-tuned for dysfluent speech
- Dual-stage dysfluency detection
- (word-level + phoneme-level)
- API-based architecture for seamless integration with EdTech & apps

Current State:

Two-level detection framework mappedgive me

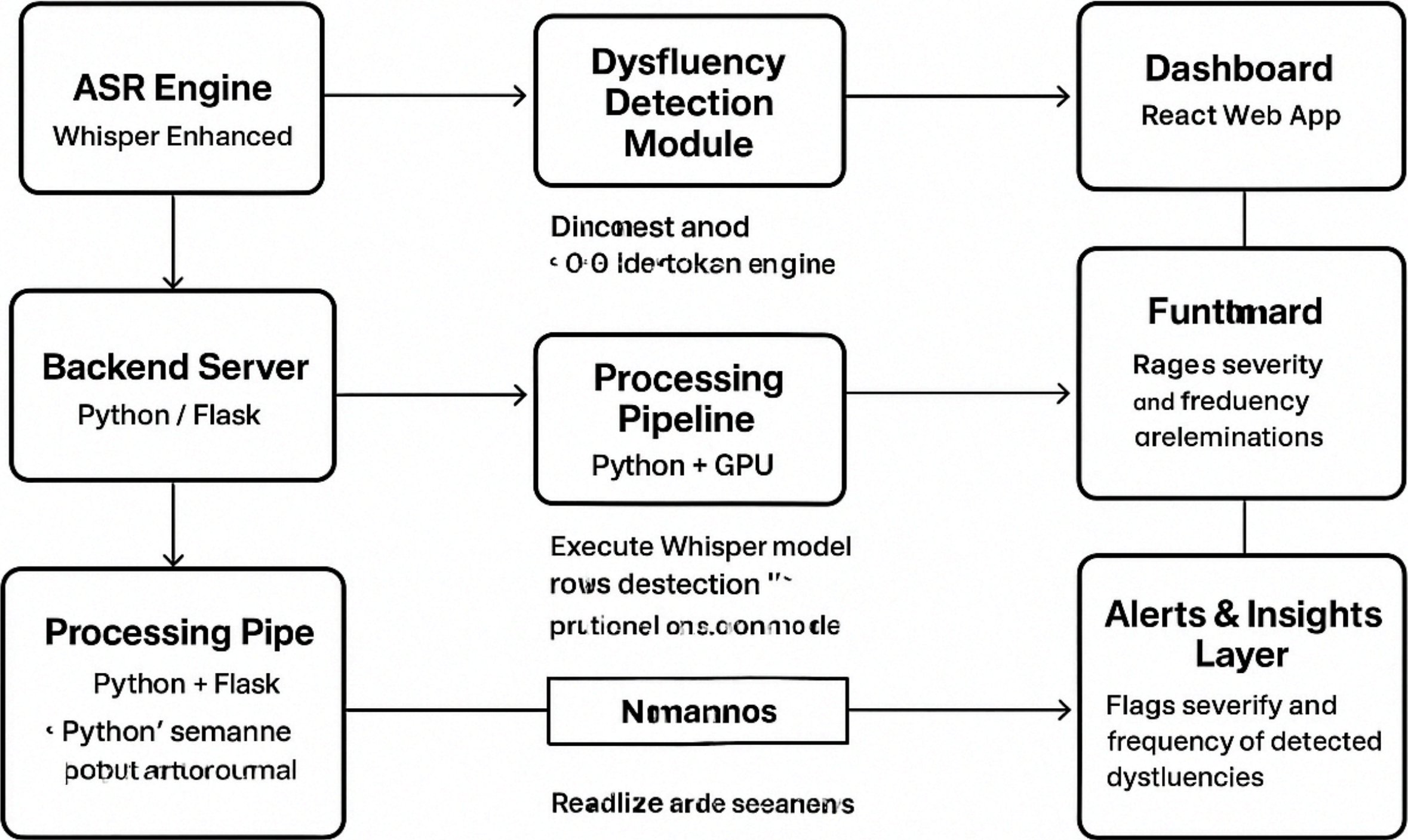
Innovation:

- Simultaneous Transcription + Detection
- Phoneme-level breakdown provides high-precision detection
- Fully modular design for integration into educational and therapy platforms

Patent filed for
**“Unmasking Silence: Advancing
Dysfluent Speech Understanding
with Next-Generation Tools
(202541046770)**

6. Technology

Prototype Block Diagram



7. Competitive Landscape

What are the alternative solutions available in the market?

Competitors / Alternatives:

- Standard ASR systems (Google Speech, Siri, Alexa)
- General-purpose NLP models without dysfluency support

Barriers to Entry:

- Need for large, high-quality dysfluent speech datasets

Our Advantage:

- Dual-stage detection: word-level + phoneme-level
- Custom dataset tailored for stutter patterns
- Can plug directly into EdTech, therapy apps, or voice assistants
- Simultaneous detection + transcription

8. Business Model

What is our revenue model?

Revenue Streams:

- ASR + Dysfluency Detection API subscription
- Licensing the inclusive ASR model
- Premium analytics dashboard for dysfluency monitoring
- Dataset licensing

Why it is viable:

A specialized ASR designed for dysfluent speakers fills a critical accessibility gap. Institutions such as schools, therapy centers, and EdTech companies benefit from a recurring subscription model

9. Traction

Customer Validation

Current Traction:

- Literature review of ASR models for dysfluency completed
- Whisper model architecture analyzed for fine-tuning
- Detection framework (word-level + phoneme-level) designed
- Initial dataset collection and preprocessing started

KPIs Achieved:

- Accurate baseline transcription using Whisper for dysfluent audio
- Successful mapping of stutter patterns for detection pipeline
- Framework for dual-stage detection (word + phoneme) finalized

10. Resource Mobilisation Plan

What is our Resource Mobilization Strategy and What we have done so far?

What we have done:

- Completed analysis of existing ASR systems (Whisper, Wav2Vec, etc.)
- Collected initial sample dataset for stutter patterns
- Created framework for phoneme-level segmentation
- Patent filed

What we need (funding use):

- Larger dysfluent speech dataset
- professional annotations
- Speech therapist collaboration for validation
- Cloud deployment & model hosting
- User testing with schools and therapy centers

Phase 1 (0–6 months):

Finalize dataset collection, prepare annotations, begin Whisper fine-tuning, and validate dual-stage detection on small test samples.

Phase 2 (6–12 months):

Improve transcription accuracy, strengthen detection pipeline, build real-time dashboard

Phase 3 (12–18 months):

Conduct large-scale validation with speech therapists and clinics

11. Way Forward Strategy

What is our Resource Mobilization Strategy and What we have done so far?

Short Term (0–6 months):

- Expand dysfluent speech dataset
- Collect professionally annotated samples with speech therapists
- Deploy cloud backend + basic web dashboard for early testers

Mid Term (6–18 months):

- Field testing with therapy centers, schools, clinics
- Pilot deployments in 1–2 therapy institutions
- Add multi-language support

Long Term (18–36 months):

- Launch commercial/clinic-ready version of the platform
- Scale to 10–20 partner therapy centers
- Collaborate with government bodies for accessible speech-therapy tech